* Learning :-

→ Find best set of parameter θ^* from the parameter space Θ

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \, L(\theta, D^{train})$$

Loss function → to measure goodness of model.

$$L(\theta, D^{train}) = \frac{1}{N} \sum_{i}^{|D^{train}|} L(\theta, (x^{(i)}, y^{(i)}))$$

0-1 loss → Loss computed as the no. of misclassification

$$L(\theta, (x, y)) = |\text{misclassification}|$$

Root mean square Error :-

→ direction comparism b/w the prediction & target label

$$L(\theta, (x, y)) = \| f(x; \theta) - y^{(i)} \|_2$$

* Gradient Descent :-

① Classify all training data to obtain the prediction $y^{(i)} = f(x_i; \Theta^{(e-1)})$ and the loss $L(\Theta^{(e-1)}, D^{train})$.

② compute the gradient

$$\Delta L = \frac{\partial L}{\partial \Theta}(\Theta^{(e-1)}, D^{train}).$$

③ update the parameter $\Theta^{(e)} = \Theta^{(e-1)} - lr \cdot \nabla L$.

* stochastic Gradient descent

→ Reduce the computational cost of gradient descent by computing the gradient of single sample.

① Shuffle the training data D^{train}.

② for $i = \ldots N-1$:

ⓐ classify x to obtain $y = f(x; \Theta^{(e*N+i)})$ and the loss $L(\Theta^{(e*N+i)}, X, y)$.

ⓑ <u>compute</u> the gradient

$$\nabla L = \frac{\partial L}{\partial \Theta}(\Theta^{(e*N+i)}, x^{(i)}, y^{(i)}).$$

ⓒ update the parameter

$$\Theta^{(e*N+i+1)} = \Theta^{(e*N+i)} - lr \cdot \nabla L$$

* Shallow neural network

-> Consist of one hidden layer b/w the i/p and output layers. It performs a sequence of linear transformation followed by a non linear activation function to learn pattern from i/p data.

$$f(x, \theta) = W_2 h + b_2$$
$$= W_2 \phi(W_1 x + b_1) + b_2 = S$$

Here, x is the i/p vector

$\theta = (W_1, b_1, W_2, b_2)$ represent all trainable parameter.

$\phi \Rightarrow$ Non linear activation (like ReLU, Sigmoid, tanh)